Run report: plain S_5 C_4 with solver options, 2026-10-19

Descriptive record of the runs behind `../../README.md`. Observations only.

- **Runner:** `examples/gaudry_quartic_c4_options.rs` (summaries: `examples/gaudry_quartic_c4_options_summary.rs`), built from `origin/main` `cb2ec6873` plus this PR's `PlainOptions` wrapper, `cargo build --release`, rustc 1.93.1.
- **Host:** Apple M4 Pro, 14 logical cores, 48 GiB, macOS 25.6 arm64; `RAYON_NUM_THREADS=4`; **no isolation tier** (no core pinning on macOS; load average about 26 from other sessions' jobs at launch). Counted `F_p` multiplications are contention-independent; wall times are contended and not evidence.
- **Instances:** `p = 269`, seed 1 (§11.17′s curve, 149-point base, residual stream advanced by the committed arm's solve).

run (UTC start)	arms	residuals	solves requested / completed	failures	record
18:51Z	committed, sparsest, sparsest+chains, sparsest+chains+replay (Latin square)	0–3	16 / 16	none	`raw/s5_p269_seed1.json`, `.log`
18:52Z	committed, sparsest, sparsest+chains (generic `d` = 4–7)	—	12 / 12	none	`raw/generic_d4-7.json`, `.log`
19:21Z	sparsest+chains+replay, all residuals constructed	0–3	4 / 4	none	`raw/s5_p269_seed1_constructed4_replay.j`
19:22Z	committed, sparsest+chains, sparsest+chains+replay (rotated)	0–12	39 / 39	none	`raw/s5_p269_seed1_r13.json`, `.log`

A pilot of the generic panel (`d` = 4–6, 18:48Z, `raw/generic_d4-6.json`) gave the same counts.

Recorded values (`F_p` multiplications per solve; `raw/summary_r13.json`, `raw/summary.json`):

run	arm	n	mean C_4	range	committed / arm (ratio of means)	oracle-equal
r13	committed	13	1.2093e12	1.1933e12–1.2270e12	1	13/13
r13	sparsest+chains	13	1.6750e11	1.6525e11–1.6895e11	7.22	13/13
r13	sparsest+chains+replay (first solve excluded)	12	8.690e10	7.893e10–1.6562e11	13.90	13/13
4-arm	sparsest	4	3.0865e11	2.919e11–3.242e11	3.93	4/4
constructed	sparsest+chains+replay (3 replays)	3	7.959e10	7.942e10–7.974e10	—	4/4, planted found 4/4

Replay outcomes in r13: 1 recorded, 11 replayed, 1 fell back (residual 3). Residuals with a nonempty decomposition: 0 (constructed) and 6. Residuals 0–3 have identical counts in the 4-arm and r13 runs, and the committed arm equals `experiments/24_gaudry_quartic_c4.json` on all four.

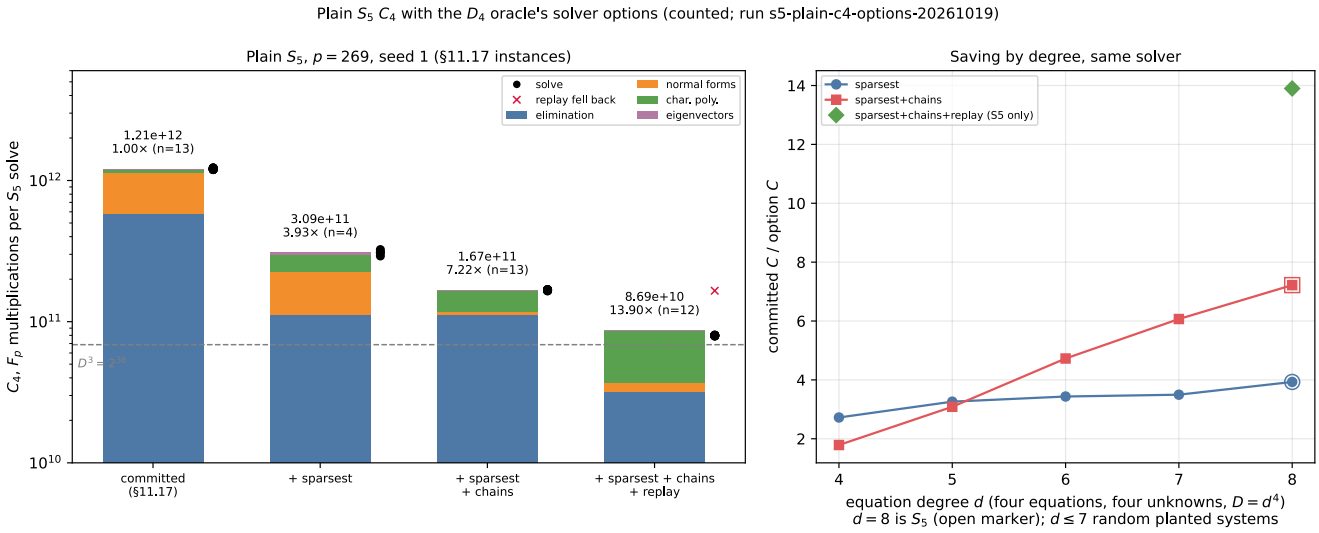


Figure 1: C4 by option

Figure: mean C_4 by arm with phase stacks and per-residual solves (left); committed/option ratio against equation degree (right). Runs above.